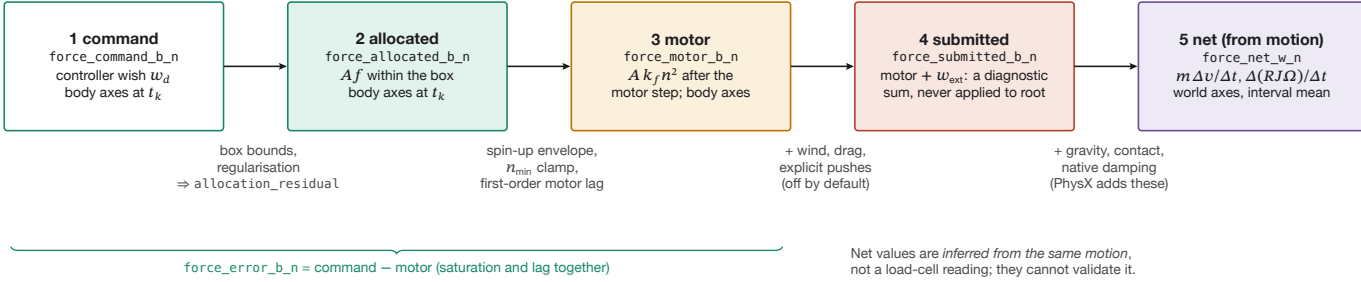
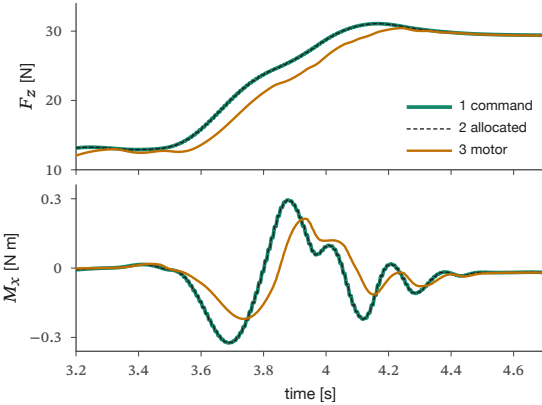


a Five force/torque stages are logged separately telemetry/measurements.py; prefixes force_*/torque_*

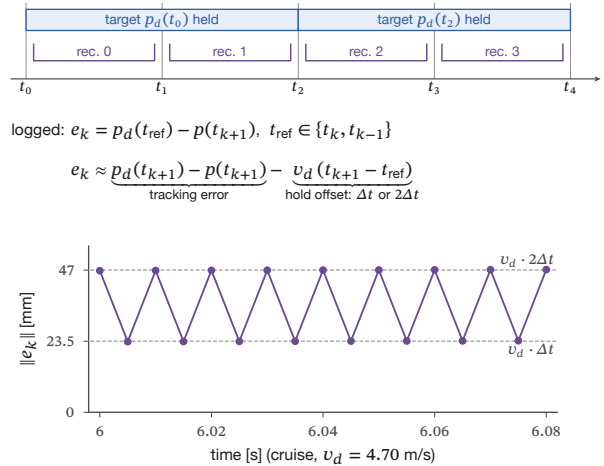


b helix acceleration ramp, synthetic plant



No bound is active here, so 1 and 2 coincide; 3 trails 1 by the motor lag ($\tau_f = 50\text{--}80$ ms). This gap is what the planner's $f + \tau_f$ margin and the feedback reserve have to absorb.

c What position_error_w_m measures



Real synthetic-run records: the logged norm alternates between 23.5 and 47.0 mm, while $\|p_d(t_{k+1}) - p(t_{k+1})\| \leq 1.0$ mm here. At speed, the hold offset dominates; phase metrics (max/RMS) include it. This is the documented definition (target actually commanded), not a bug.

d Every tracked quantity is logged as ideal, actual and error = ideal – actual basic.csv / basic

quantity	ideal	actual	note
position, velocity	held trajectory target	post-step CoM truth	world frame; norms logged too
acceleration	held target a_d	$\Delta v / \Delta t$ over the step	includes gravity's effect; not IMU specific force
attitude	controller R_d (held)	post-step R	Z-Y-X angles for reading; use attitude_error_angle_rad
angular velocity	$R^T R_d \Omega_d$ in actual body axes	Ω_b	PID: its rate set-point
motor speed	RPS command after limits	integrated motor state	rps, rpm, rad/s columns
rotor thrust	$k_f n_{cmd}^2$	$k_f n^2$	per rotor BL, BR, FL, FR
body force / torque	stage 1 command	stage 3 motor	error = saturation + lag